

Geometric Dosimetry

Closed-form absorption laws
from 100 MHz to 100 GHz

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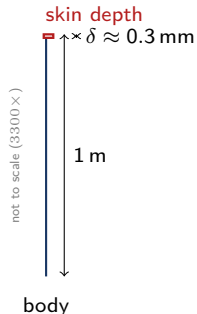
Problem: FDTD at mmWave is intractable

The scale mismatch:

- Body span: ~ 1 m
- Skin depth at 28 GHz: $\delta \approx 0.3$ mm
- Five orders of magnitude

FDTD consequence:

- Mesh: $\sim 10^{12}$ cells per simulation
- One plane wave, one phantom, one frequency:
weeks
- Systematic campaign (550 runs): months of
GPU



We replace the volumetric simulation with surface-level geometric operations.

Approach in one slide

1. Exact local law from Fresnel + Poynting at every surface point:

$$S_{\text{ab}}(\mathbf{r}) = S_{\text{inc}} T_{\text{eff}}(\mathbf{r}) \text{ReLU}[\hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}})]$$

2. Pseudo-Brewster compensation: for biological tissue, the Fresnel transmission is nearly angle-independent. Material collapses to a scalar T_0 :

$$S_{\text{ab}}(\mathbf{r}) \approx S_{\text{inc}} T_0 \text{ReLU}[\hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}})]$$

3. All variation is geometry: projected area, ambient occlusion, Cauchy's formula. Dosimetry chain = single-hidden-layer ReLU network. Differentiable.

$10^6 \times$ speedup | 5–15 % error | within tissue data uncertainty
(~20 %)

Structure

Part I

mmWave Framework

- Exact absorption law
- Pseudo-Brewster
- Geometric framework
- Compliance
- Computation
- Validation

Part II

Generalisations

- Polarisation-aware
- Stokes vector
- Multipath convergence
- Sub-6 GHz extension
- Exact $\bar{T}$

Part III

Coherent MIMO

- Coherent absorption law
- Exposure operator $\mathbf{Q}$
- MRT analysis
- Hotspot formation
- Exposure-constrained BF
- Multi-user MIMO

Literature review

	$T_{\text{avg}} \approx T_0$	Local $S_{\text{ab}}(\mathbf{r})$	Cauchy $A_{\text{ab}}/4$	$\eta = \text{AO}$	Compl. formula	Expos. op. $\mathbf{Q}$
Kodera 2024	○	—	○	—	—	—
Funahashi 2018	—	△	—	—	—	—
Li 2019	○	△	—	—	—	—
Flintoft 2014	—	—	○	—	—	—
Zhang 2020	○	—	○	—	—	—
Azzam 2015	●	—	—	—	—	—
Ying 2015, 2017	—	—	—	—	○	●
Castellanos 2020	—	△	—	—	—	△
This work	●	●	●	●	●	●

● = derived analytically ○ = observed empirically △ = partial — = not addressed

► Full literature tables

Exact absorption law

Three factors determine local absorption:

- Illumination: incident power density S_{inc}
- Material: effective Fresnel transmission $T_{\text{eff}}(\mathbf{r})$
- Geometry: visibility gate $\text{ReLU}(\hat{\mathbf{n}} \cdot (-\hat{\mathbf{k}}))$

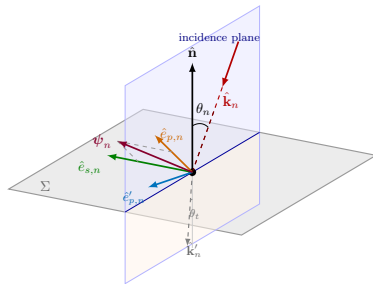
Exact absorption law

$$S_{\text{ab}}(\mathbf{r}) = S_{\text{inc}} T_{\text{eff}}(\mathbf{r}) \text{ReLU}[\hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}})]$$

$T_{\text{eff}} = |e_s|^2 T_s(\theta) + |e_p|^2 T_p(\theta)$: Fresnel-weighted average.
ReLU gate is exact: back-facing points absorb nothing.

► Fresnel derivation

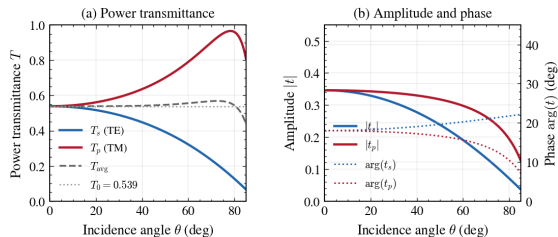
► Exact law details



Pseudo-Brewster compensation

T_s decreases with angle; T_p increases.
Average $T_{\text{avg}}(\theta)$ stays near T_0 .

Tissue	$ \tilde{n} $	T_0	T_{avg}/T_0 var.
Skin	4.84	0.54	$\leq 5.6\%$
Muscle	5.62	0.48	$\leq 4.8\%$
Fat	2.05	0.77	$\leq 8.2\%$
Water	6.62	0.45	$\leq 3.9\%$



Geometric absorption law

$$S_{\text{ab}}(\mathbf{r}) \approx S_{\text{inc}} T_0 \text{ReLU}[\hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}})]$$

Azzam (2015): holds for $|\tilde{n}| > 2.5$.

Biological tissue: $|\tilde{n}| \approx 3-6$. ✓

► Pseudo-Brewster details

Geometric consequences

Total absorbed power: $P_{\text{abs}} = S_{\text{inc}} T_0 A_{\perp}(\hat{\mathbf{k}})$
 ($A_{\perp}$ = shadow area)

Self-shadowing on non-convex bodies:

$$\eta(\mathbf{r}) = \frac{1}{\pi} \int_{S^2} \text{ReLU}[\hat{\mathbf{n}} \cdot (-\hat{\mathbf{k}})] O(\mathbf{r}, \hat{\mathbf{k}}) d\Omega$$

$\equiv$ ambient occlusion (GPU graphics);

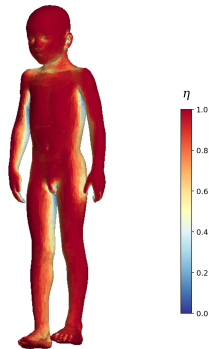
Thelonious: $A_{\text{ab}}/A \approx 0.87$.

Generalised Cauchy formula

$$\langle P_{\text{abs}} \rangle = S_{\text{inc}} T_0 \frac{A_{\text{ab}}}{4}$$

Classical Cauchy ($\langle A_{\perp} \rangle = A/4$ for convex) is special case $\eta \equiv 1$.

► Cauchy proof & ambient occlusion



Exposure fraction $\eta(\mathbf{r})$ on Thelonious.
 Red = exposed, blue = self-shadowed.

Absorption directivity

Factored result:

$$P_{\text{abs}}(\hat{\mathbf{k}}) = S_{\text{inc}} T_0 \frac{A_{\text{ab}}}{4} D(\hat{\mathbf{k}})$$

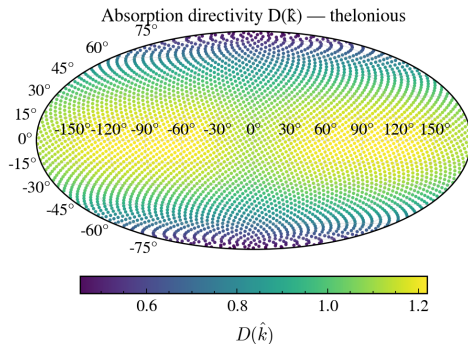
Separation of concerns:

- EM environment: S_{inc}
- Tissue: T_0 (one number)
- Body geometry: A_{ab} and $D(\hat{\mathbf{k}})$, both precomputed

$D(\hat{\mathbf{k}})$ is smooth on S^2 :

SH expansion with $L \leq 6$ (49 coeff.)

gives RMS error $< 1\%$.



Absorption directivity $D(\hat{\mathbf{k}})$,
Mollweide projection. Range: 0.8–2.0.

Computational framework

Multi-source environment: N paths with directions $\mathbf{K}$, power densities $\mathbf{s}$:

Dosimetry as a ReLU neural network

$$\mathbf{S}_{\text{ab}} = T_0 \text{ReLU}(\mathbf{N}\mathbf{K}^T) \mathbf{s}$$

- $\mathbf{N}$: surface normals (body mesh)
- $\mathbf{K}$: path directions (ray tracer)
- $\mathbf{s}$: path power densities (ray tracer)
- ReLU: visibility gating (exact)

Differentiable: gradients of SAR w.r.t. antenna positions propagate by backpropagation through any ray-tracing engine.

Point source ($d > 3\lambda$):

$$P_{\text{abs}} = P_t T_0 \frac{\Omega_{\text{body}}}{4\pi}$$

Emissive view-factor formula of radiative heat transfer, with T_0 replacing emissivity.

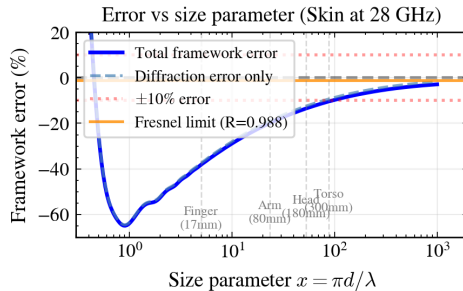
Validation: Mie theory and phantom

Mie theory (exact for lossy spheres):

- Tests Fresnel + geometric optics jointly
- Body-scale objects at mmWave: $\sim 10\%$ error
- Dominated by diffraction into geometric shadow
- Error is conservative (underestimates)

Thelonious phantom:

- Total absorbed power error: 0.35%
- Local RMS: 3.2%
- Isolates the Fresnel approximation from diffraction



(a) Error vs size parameter at 28 GHz.
Body parts annotated by diameter.

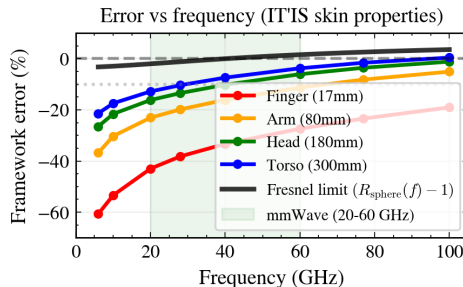
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(b) Error vs frequency for different body parts.
Torso at 28 GHz: $\sim 5\%$.

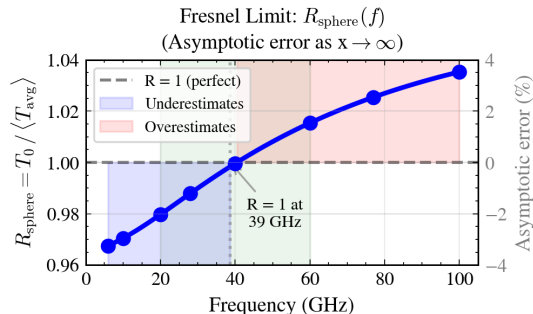
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(c) Fresnel asymptotic limit $R_{\text{sphere}}(f)$.
Conservative below 40 GHz.

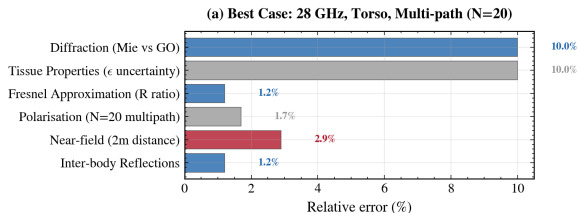
Error budget

Source	Magnitude
Fresnel (local)	$\sim 5\%$
Fresnel (integrated)	$\sim 1\%$
Diffraction	$\sim 10\%^*$
Curvature	$\sim 0.5\%$
Tissue uncertainty	$\sim 20\%$
Total framework	10–15 %

*Worst case for body parts $< 10\lambda$.

Conservative (underestimates absorption).

Framework error $<$ tissue data uncertainty.



(b) Worst Case: 6 GHz, Finger, Single-path LOS

(a) Best case: 28 GHz, torso, multipath ($N=20$).

All errors $\leq 10\%$; dominated by diffraction and tissue.

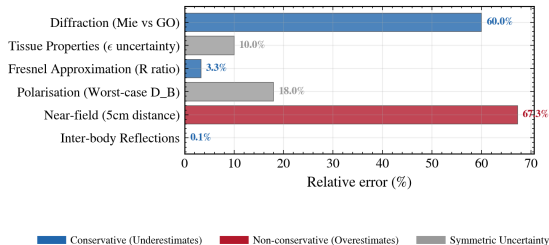
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Total framework	10–15%

*Worst case for body parts $< 10\lambda$.

Conservative (underestimates absorption).

Framework error $<$ tissue data uncertainty.



(b) Worst case: 6 GHz, finger, single-path LOS.
Near-field and diffraction dominate.

Part I: detail index

▸ Setup and notation

▸ Power flux derivation

▸ Fresnel coefficients

▸ Exact polarisation-aware law

▸ Pseudo-Brewster mechanism

▸ Projected area and Cauchy

▸ Absorption directivity

▸ Whole-body SAR compliance

▸ Computational framework

▸ Higher-order corrections

▸ Mie validation details

▸ 3D absorbed power density

▸ Posture sensitivity

▸ Near-field extension

▸ Inter-body reflections

Part II: relaxing two assumptions

Part I assumed:

1. Unpolarised or circular illumination ($q = 0$)
2. mmWave frequencies ($T_{\text{avg}} \approx T_0$)

Part II retains both exactly:

1. Arbitrary polarisation
→ absorption Stokes vector
2. Any frequency above ~ 100 MHz
→ exact flux-averaged $\bar{T}$

Key results:

- Worst-case polarisation correction on human body: 16 %
- Universal bound (cylinder): 27.9 %
- With ≥ 20 multipath: < 2.5 %
- $\bar{T}$ makes Cauchy formula exact at any frequency

Absorption Stokes vector

Retain polarisation exactly. Integrate over body surface:

Stokes vector representation

$$P_{\text{abs}} = \mathbf{m}(\hat{\mathbf{k}}) \cdot \mathbf{s}_{\text{inc}}$$

$\mathbf{s}_{\text{inc}} = (S_0, S_1, S_2, S_3)^\top$: incident Stokes vector

$$\mathbf{m}(\hat{\mathbf{k}}) = (\tilde{A}_\perp, -\frac{1}{2}B_c, -\frac{1}{2}B_s, 0)^\top$$

- Both $\tilde{A}_\perp$ and $\tilde{B}$ precomputed from mesh
- Fourth component = 0: circular handedness irrelevant

► Stokes derivation

Polarisation directivity D_B :

- Bounds fractional correction
- Cylinder (worst case): 27.9 %
- Thelonious (max over S^2): 16.0 %

Unpolarised geometric law is conservative for vertical polarisation: horizontal limbs are purely TE, $T_s \leq T_0$.

► D_B bounds

► Canonical shapes

Multipath convergence

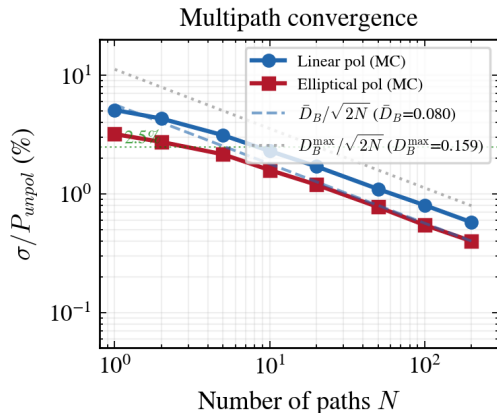
With N independent paths, polarisation errors cancel:

$$\frac{\sigma}{P_{\text{abs}}} = \frac{D_B}{\sqrt{2N}}$$

N paths	Pol. error (std)
1	up to 16 %
10	≤ 3.6 %
20	≤ 2.5 %
50	≤ 1.6 %

In realistic indoor mmWave channels ($N \geq 20$), polarisation error is below the pseudo-Brewster error.

► Convergence details



(a) Convergence of polarisation error with N .

$D_B/\sqrt{2N}$ bound (dashed).

Multipath convergence

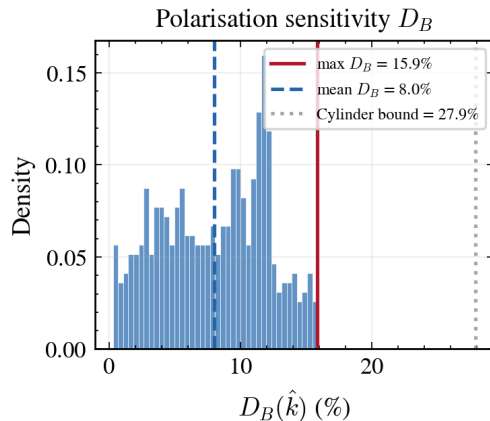
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► Convergence details



(b) Histogram of D_B over directions.

Max 16 %, mean 8 %, cylinder bound 27.9 %.

Sub-6 GHz extension

$T_{\text{avg}}(\theta) \approx T_0$ weakens below 10 GHz. Replace T_0 with the exact flux-averaged transmission:

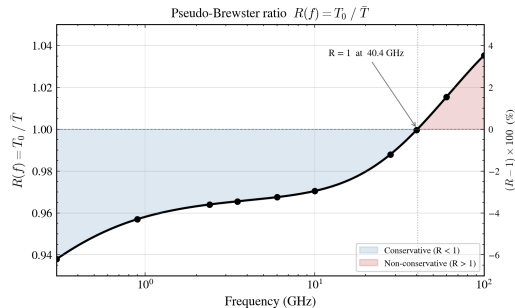
$$\bar{T}(f) = 2 \int_0^1 T_{\text{avg}}(\mu, f) \mu \, d\mu$$

Exact at any frequency

$$\langle P_{\text{abs}} \rangle = S_{\text{inc}} \bar{T} \frac{A_{\text{ab}}}{4}$$

Below 40 GHz: T_0 underestimates (conservative).

Below 6 GHz: local map loses meaning; total power via $\bar{T}$ remains valid.



(a) Pseudo-Brewster ratio $R(f) = T_0 / \bar{T}$.
Conservative ($R < 1$) below 40 GHz.

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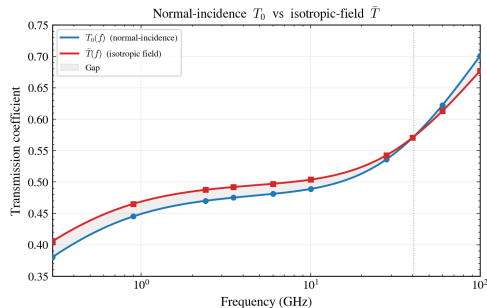
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(b) T_0 vs $\bar{T}$ across frequency.

Gap widens below 6 GHz.

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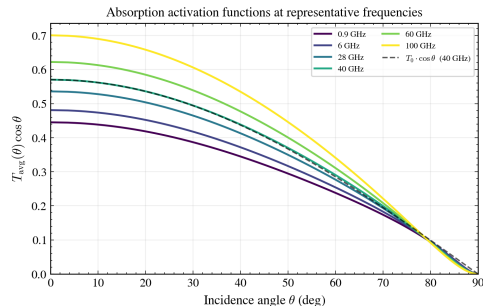
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(c) $T_{\text{avg}}(\theta) \cos \theta$ at representative frequencies.

► $\bar{T}$ derivation

Part II: detail index

▸ Stokes vector derivation

▸ Polarisation sensitivity D_B

▸ Canonical shapes (sphere, cylinder)

▸ Polarisation on human body

▸ Multipath convergence details

▸ SH compression of polarisation

▸ Exact averaged transmission $\bar{T}$

▸ Validity regimes

Part III: from incoherent to coherent

Parts I–II: incoherent. Powers add.

Part III: coherent MIMO. Fields add.

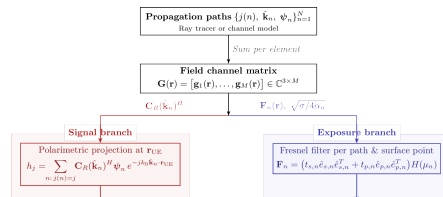
M antenna elements, precoding vector $\mathbf{x} \in \mathbb{C}^M$:

- Coherent peak can exceed incoherent sum by factor N
- Part I–II framework cannot capture this

Two branches diverge:

- Signal at UE: sums *scalars* $\rightarrow \mathbf{h}^\top \mathbf{x}$
- Exposure on body: sums *vectors* $\rightarrow \tilde{\mathbf{G}}(\mathbf{r}) \mathbf{x}$

Exact complex Fresnel amplitudes t_s, t_p must be retained; replacing by $\sqrt{T_0}$ destroys cross-term phases.



(a) Propagation paths $\rightarrow$ field channel matrix $\rightarrow$ branch split.

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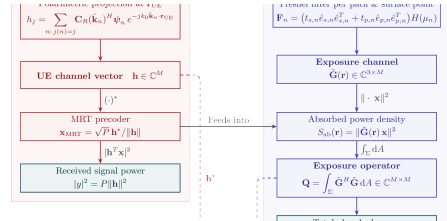
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(b) Signal branch (scalar) vs exposure branch (vector).

Part III: from incoherent to coherent

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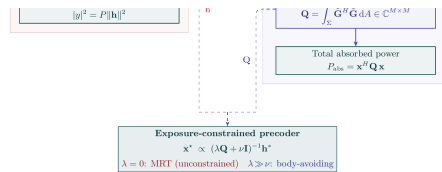
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Exact complex Fresnel amplitudes t_s, t_p must be retained; replacing by $\sqrt{T_0}$ destroys cross-term phases.



$\mathbf{F}_n^H$ collapses each path to a scalar at $\mathbf{r}_{\text{UE}}$.
 $\mathbf{F}_n(\mathbf{r})$ retains the full vector structure over the body surface Σ .

(c) Outputs $\rightarrow$ exposure-constrained precoder.

► Full diagram

Coherent absorption law

Absorbed power density = Ohmic loss integrated over depth.

Cross-terms weighted by depth coupling factors $\Lambda_{nn'}$.

Two approximations collapse double sum to squared norm:

1. **TM direction:** replace refracted by incident.

Error: $O(1/|\tilde{n}|^2) \lesssim 4\%$ on cross-terms.

Self-terms: exact.

2. **Depth coupling:** set $\Gamma_{nn'} = 1$.

Exact to 0.44 % at 28 GHz.

Both affect only cross-terms.

Combined error $\lesssim 5\%$.

► Depth integral

► Approx. 2

Coherent absorption law

$$S_{\text{ab}}(\mathbf{r}) = \|\tilde{\mathbf{G}}(\mathbf{r}) \mathbf{x}\|^2$$

$\tilde{\mathbf{G}}(\mathbf{r}) \in \mathbb{C}^{3 \times M}$: exposure channel matrix.

Collects Fresnel-filtered, depth-weighted field contributions.

When phases are random, cross-terms average to zero
 $\rightarrow$ reduces to Part I multi-source formula.

Exposure operator

Integrate coherent law over body surface:

Exposure operator

$$P_{\text{abs}} = \mathbf{x}^H \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} \equiv \int_{\Sigma} \tilde{\mathbf{G}}(\mathbf{r})^H \tilde{\mathbf{G}}(\mathbf{r}) dA \in \mathbb{C}^{M \times M}$$

Properties:

- Hermitian positive-semidefinite
- Eigendecomposition: $\mathbf{Q} = \sum_k \lambda_k \mathbf{v}_k \mathbf{v}_k^H$
- $\mathbf{v}_1$ maximises body absorption
- $\lambda_{\max}$: worst-case absorption per unit power
- $\text{tr}(\mathbf{Q})$: expected absorption under isotropic TX

Analogy with Part I:

Part I	Part III
$D(\hat{\mathbf{k}})$	$\lambda_k, \mathbf{v}_k$
Direction	Precoding vector
$D_{\max} = 2$	$\lambda_{\max}$
Precomputed	Precomputed

MRT and exposure-signal alignment

Under maximum-ratio transmission:

$$P_{\text{abs}}^{\text{MRT}} = P \rho \lambda_{\max}(\mathbf{Q})$$

Exposure-signal alignment

$$\rho \equiv \frac{\mathbf{h}^{\top} \mathbf{Q} \mathbf{h}^*}{\|\mathbf{h}\|^2 \lambda_{\max}(\mathbf{Q})} \in [0, 1]$$

Three reasons signal focus $\neq$ body hotspot:

1. Polarisation

Transmitted polarisations partially cancel at body surface.

Small ρ : body geometry gives inherent protection under MRT.

2. Grazing paths

Strong signal contribution, weak absorption ($\cos \theta \rightarrow 0$).

3. Phase divergence

Extra per-path phases accumulate between UE and body surface.

Coherent hotspot formation

Hotspot forms when paths are:

- Co-phased at a body-surface point
- Co-polarised after Fresnel filtering
- Arriving from a narrow angular spread

Hotspot size:

$$\Delta r \sim \frac{\lambda}{\Delta \theta}$$

Scales with wavelength and inversely with angular spread of contributing paths.

Rich scattering (many directions) $\rightarrow$ smaller hotspot. Sparse channel (few directions) $\rightarrow$ broader, weaker hotspot.

Fresnel filtering:

- Body's transmission operator acts as spatial filter
- Grazing-incidence paths suppressed by $\cos \theta$ gate
- N^2 peak scaling (coherent) vs N (incoherent)

Exposure-constrained beamforming

Problem: maximise signal subject to absorption limit:

$$\max_{\mathbf{x}} |\mathbf{h}^\top \mathbf{x}|^2 \quad \text{s.t.} \quad \mathbf{x}^H \mathbf{Q} \mathbf{x} \leq P_{\text{abs}}^{\max}, \quad \|\mathbf{x}\|^2 \leq P$$

S-procedure guarantees global closed-form solution:

Optimal precoder

$$\mathbf{x}^* = \sqrt{P} \frac{(\lambda \mathbf{Q} + \nu \mathbf{I})^{-1} \mathbf{h}^*}{\|(\lambda \mathbf{Q} + \nu \mathbf{I})^{-1} \mathbf{h}^*\|}$$

Three limiting regimes:

$\lambda = 0$:

MRT (unconstrained).

Full signal, no exposure limit.

$\lambda \gg \nu$:

Body-avoiding precoder.

Suppresses $\mathbf{v}_1$ direction.

$\rho \ll 1$:

Natural protection.

Minimal signal loss.

Same structure as Ying *et al.* 2015, but $\mathbf{Q}$ derived analytically from Fresnel physics, not FDTD-calibrated.

Multi-user MIMO

K users, each with their own body and exposure operator:

Per-person exposure:

$$P_{\text{abs}}^{(u)} = \mathbf{x}^H \mathbf{Q}^{(u)} \mathbf{x}$$

$\mathbf{Q}^{(u)}$ built from paths arriving at user u 's body.

Per-person constraints:

$$\mathbf{x}^H \mathbf{Q}^{(u)} \mathbf{x} \leq P_{\text{abs}}^{\max}, \quad u = 1, \dots, K$$

Integration with standard solvers:

- Additional quadratic constraints in WMMSE
- No new approximations beyond single-user
- Additive structure: $\mathbf{Q}^{(u)}$ computed independently per user

Same $10^6\times$ speedup preserved in the coherent pipeline.

Part III: detail index

▸ Setup and assumptions

▸ Propagation paths

▸ Field channel matrix

▸ Signal vs exposure branch

▸ Fresnel transmission operator

▸ Depth integral and Approx. 1

▸ Depth coupling and Approx. 2

▸ Main theorem

▸ Exposure operator properties

▸ Exposure-signal alignment ρ

▸ QCQP solution details

▸ Computational cost

Summary of results

Part I: mmWave

$$S_{\text{ab}} \approx S_{\text{inc}} T_0 \text{ReLU}(\mu)$$

$10^6 \times$ speedup

5–15 % error

$$\langle P_{\text{abs}} \rangle = S_{\text{inc}} T_0 A_{\text{ab}} / 4$$

Compliance:

$$S_{\text{inc}} < 0.16 \text{ m} / (T_0 A_{\text{ab}})$$

ReLU network formulation

Part II: Generalisations

$$P_{\text{abs}} = \mathbf{m}(\hat{\mathbf{k}}) \cdot \mathbf{s}_{\text{inc}}$$

Polarisation: $\leq 16\%$ on body

Multipath ($N \geq 20$): $\leq 2.5\%$

$$\langle P_{\text{abs}} \rangle = S_{\text{inc}} \bar{T} A_{\text{ab}} / 4$$

Exact 100 MHz–100 GHz

Part III: Coherent MIMO

$$S_{\text{ab}} = \|\tilde{\mathbf{G}} \mathbf{x}\|^2$$

Exposure operator

$$\mathbf{Q} = \int \tilde{\mathbf{G}}^H \tilde{\mathbf{G}} \text{d}A$$

Alignment $\rho \in [0, 1]$

$$\text{QCQP: } \mathbf{x}^* \propto (\lambda \mathbf{Q} + \nu \mathbf{I})^{-1} \mathbf{h}^*$$

Approx. error $\lesssim 5\%$

Five irreducible limitations: reactive near-field ($d < \lambda/2\pi$) | coherent multi-bounce | diffraction at edges |
volumetric SAR below 6 GHz | tissue dielectric uncertainty ($\sim 20\%$)

Outlook and dissemination

Publication plan:

1. **arXiv** (priority claim)
Post full monograph. Establishes priority on all results.
2. **Paper A** → IEEE TAP or T-EMC
Base framework (Parts I–II).
Mostly condensation from the monograph.
3. **Paper B** → IEEE TWC or JSAC
Coherent extension (Part III).
Needs simulation study: Sionna RT + Thelonious phantom.
4. **BioEM 2027** abstract
Reuse existing draft. Part I focus.

Framing strategy:

Paper A opening:

“Kodera *et al.* (2024) observed numerically that ACS converges to projected area above 10 GHz. We derive this analytically, identify pseudo-Brewster as the physical origin, and extend to local fields, arbitrary polarisation, diffuse fields, and sub-GHz frequencies.”

Missing piece for Paper B:

Sionna RT simulation study:
64-element BS, 28 GHz,
MRT vs optimal QCQP,
hotspot intensity and SNR penalty.

Thank you

Questions and discussion

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Prior art comparison (full table)

	$T_{\text{avg}} \approx T_0$	Local $S_{\text{ab}}(\mathbf{r})$	Cauchy $A_{\text{ab}}/4$	η = AO	Compl. formula	Expos. op. $\mathbf{Q}$
Kodera 2024	○	—	○	—	—	—
Funahashi 2018	—	△	—	—	—	—
Li 2019	○	△	—	—	—	—
Samaras & Kuster 19	△	—	—	—	—	—
Diao 2021	○	—	—	—	—	—
Diao 2024	○	—	—	—	—	—
Sasaki 2017	△	—	—	—	—	—
Bamba 2012, 2014	○	—	—	—	—	—
Flintoft 2014	—	—	○	—	—	—
Zhang & Robinson 20	○	—	○	—	—	—
Azzam 2015	●	—	—	—	—	—
Gosselin 2011	—	—	—	—	○	—
Hochwald 2014	—	—	—	—	—	○
Ying 2015, 2017	—	—	—	—	○	●
Castellanos 2020	—	△	—	—	—	△
This work	●	●	●	●	●	●

● = derived analytically ○ = observed empirically △ = partial — = not addressed

◀ Back to literature slide

◀ Back to summary

Framework predictions vs literature

Published observation	Framework prediction	Agr.
Bamba 2012: $\eta \approx 0.5$ at 1.5–6 GHz	$\eta = T_0$; pseudo-Brewster explains weak f -dep.	2–4 %
Kodera 2024: $ACS \propto$ projected area > 10 GHz	$T_{\text{avg}} \approx T_0 \Rightarrow P_{\text{abs}} = T_0 A_{\perp} S_{\text{inc}}$	< 5 %
Diao 2024: $T \approx 0.52$ at 28 GHz (FDTD)	Normal-incidence Fresnel: $T_0 = 0.536$	3 %
Flintoft 2014: $\langle Q^a \rangle \approx 0.41$, 7–11 GHz	$T_0 \times \eta \approx 0.48 \times 0.85 = 0.41$	3–8 %
Zhang 2020: $\xi \approx 0.45$ – 0.65 above 6 GHz	$\xi = \bar{T} \times A_{\text{ab}}/A$; predicted 0.43–0.49	Within scatter
Li 2019: flux insensitive to angle, 30 GHz–1 THz	Pseudo-Brewster: $T_{\text{avg}}/T_0 \approx 1$ for $ \tilde{n} > 2.5$	< 5.6 %
Diao 2021: heating factor $\approx \text{const}$ vs angle	TE decreases, TM increases; $T_{\text{avg}} \approx T_0$	< 6 %

◀ Back to literature slide

Prior art: local absorption and dosimetry

Key finding: all observe T_{tr} numerically; none derive the mechanism.

- **Kodera 2024:** $\text{SAR}_{\text{wb}} = T_{\text{tr}} \times A_{\perp} \times S_{\text{inc}}/W$ from 1D slab. Reproduces 3D FDTD to 5 %, 10–100 GHz. No physical derivation.
- **Funahashi 2018:** $S_{\text{inc}} \times T_{\text{tr}}$ correlates with temperature rise to 15 %, 30–300 GHz. Normal incidence only.
- **Li 2019:** transmitted flux insensitive to angle/polarisation, 6 GHz–1 THz. Unpolarised average T_{avg} not examined.
- **Samaras & Kuster 2019:** TE/TM transmittances vs angle. TM worst case 0.92 at pseudo-Brewster angle (exceeds ICNIRP by 70 %). Unpolarised average not evaluated.
- **Diao 2024:** $T \approx 0.52$ at 28 GHz from full FDTD. Fresnel: $T_0 = 0.536$ (3 % discrepancy).
- **Azzam 2015:** proved $|\tilde{n}| > 2.5 \Rightarrow \text{reflectance} \approx \text{const over } 0\text{--}60^\circ$. Optics; not applied to dosimetry.

Prior art: diffuse-field dosimetry

Key finding: Cauchy's projected area and a transmission coefficient appear as separate empirical ingredients. Not connected to Fresnel theory.

- **Andersen 2007:** room electromagnetics. $P_{\text{abs}} = S_{\text{total}} \langle \sigma_a \rangle$. 153 citing papers.
- **Bamba 2012, 2014:** whole-body SAR in reverberation chamber at 1.8 GHz. $\eta \approx 0.5$ from FDTD (1.45–5.8 GHz). No physical origin identified.
- **Flintoft 2014:** ACS of 60 volunteers, 1–12 GHz. Normalised ACS $\langle Q^a \rangle \approx 0.41$ at 7–11 GHz. Dip below 3 GHz attributed to fat-layer standing waves; no quantitative model.
- **Zhang 2017, 2020:** ACS of 48 subjects, 1–18 GHz. $\xi \approx 0.45$ –0.65 above 6 GHz. ACS–BSA correlation > 0.9 . Physical content $\xi = \bar{T} \times A_{\text{ab}}/A$ not derived.
- **Hallbjörner 2005:** ACS of large lossy objects depends only on permittivity, not shape, in skin-depth limit. Mechanism (pseudo-Brewster) not identified.

Prior art: coherent MIMO and exposure

Key finding: all prior work has FDTD-calibrated exposure matrices. No closed-form physical content.

- **Hochwald 2014:** SAR matrix $\mathbf{S}$ at 1.8 GHz. SAR-aware coding: +2.5 dB over Alamouti. $\mathbf{S}$ calibrated numerically per device.
- **Ying 2015:** capacity-optimal precoder $\mathbf{x}^* = (\lambda \mathbf{S} + \nu \mathbf{I})^{-1} \mathbf{h}^*$. Strong duality proved. SU-MIMO uplink.
- **Ying 2017:** MU-MIMO uplink. Per-user SAR matrix. Decentralised protocol.
- **Castellanos 2020:** per-element Fresnel at 28 GHz. Calibration scalar $\beta \approx 0.70$. Relation to T_0 not examined. No body-surface integral.
- **Ebadi-Shahrivar 2019:** worst-case SAR = $\lambda_{\max}(\mathbf{S})$. Compliance via eigendecomposition.
- **Zhou 2026:** pointwise exposure matrices from dipole radiation at 30 GHz + thermal model. $T_{\text{tr}} = 0.8$ as parameter (not derived).

All assume uplink. Downlink base-station geometry not addressed.

Prior art: numerical approaches and limits

FDTD remains the regulatory standard.

- Single plane wave at 28 GHz on anatomical phantom: $\sim 10^{12}$ mesh cells
- Systematic campaign: 15 frequencies $\times$ 4 phantoms $\times$ 2 scenarios = 550 simulations, weeks of GPU

Shortcuts pursued:

- **Kodera 2024**: 1D multilayer slab $\rightarrow T_{tr}$. Reproduces 3D to 5 %, 10–100 GHz.
- **Kushiya & Nagaoka 2025**: spherical-wave lookup tables (~ 2.2 TB per antenna). Field superposition without re-running FDTD.
- **Zhang 2017**: surface-integral-equation method (ABCD + MoM). Memory scales with surface area, not volume. Validated in 2D at 1–6 GHz.

All retain numerical tissue-parameter input. No closed-form transmission coefficient or exposure operator.

Setup and notation

Surface point $\mathbf{r}$ with outward normal $\hat{\mathbf{n}}(\mathbf{r})$.

Incident plane wave: power density S_{inc} ,

propagation direction $\hat{\mathbf{k}}$. **Incidence cosine:**

$$\mu(\mathbf{r}) = \hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}})$$

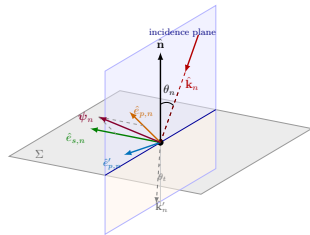
Local incidence angle: $\theta = \arccos(\mu)$.

The wave decomposes locally into:

- TE (s-polarisation): $\mathbf{E}$ perpendicular to incidence plane
- TM (p-polarisation): $\mathbf{E}$ in the incidence plane

Incidence plane defined by $\hat{\mathbf{k}}$ and $\hat{\mathbf{n}}(\mathbf{r})$.

For $\mu < 0$: back-facing, no illumination.



Power flux and absorption derivation

Starting point: Poynting's theorem at a locally flat interface.

Absorbed power density = net inward Poynting flux at the surface:

$$S_{\text{ab}}(\mathbf{r}) = -\hat{\mathbf{n}}(\mathbf{r}) \cdot \frac{1}{2} \text{Re}[\mathbf{E}_t(\mathbf{r}) \times \mathbf{H}_t^*(\mathbf{r})]$$

For a locally flat surface with Fresnel transmission:

$$S_{\text{ab}}(\mathbf{r}) = S_{\text{inc}} T_{\text{eff}}(\mathbf{r}) \cos \theta \cdot H(\mu)$$

where $H(\mu)$ is the Heaviside step function (visibility gate).

Since $\cos \theta \cdot H(\mu) = \text{ReLU}(\mu)$ exactly:

Exact absorption law

$$S_{\text{ab}}(\mathbf{r}) = S_{\text{inc}} T_{\text{eff}}(\mathbf{r}) \text{ReLU}[\hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}})]$$

The ReLU gate is exact: it encodes both the $\cos \theta$ projection and the front/back-face selection in a single operation. [◀ Back to Part I index](#)

Fresnel coefficients for lossy half-space

Complex refractive index:

$$\tilde{n} = \sqrt{\tilde{\epsilon}_r} = \sqrt{\epsilon_r - j \frac{\sigma}{\omega \epsilon_0}}$$

Complex Snell's law: $\sin \theta_t = \sin \theta_i / \tilde{n}$

Field transmission coefficients:

$$t_s = \frac{2 \cos \theta_i}{\cos \theta_i + \tilde{n} \cos \theta_t}$$

$$t_p = \frac{2 \tilde{n} \cos \theta_i}{\tilde{n} \cos \theta_i + \cos \theta_t}$$

Power absorption coefficients:

$$T_{s,p}(\theta) = \operatorname{Re} \left[\frac{\tilde{n} \cos \theta_t}{\cos \theta_i} \right] |t_{s,p}|^2$$

At 28 GHz:

Tissue	$ \tilde{n} $	T_0
Skin	4.84	0.54
Muscle	5.62	0.48
Fat	2.05	0.77
Water	6.62	0.45

Tissue data: IT'IS v5.0 database.

Exact polarisation-aware law

The effective transmission decomposes as:

$$T_{\text{eff}}(\mathbf{r}) = |e_s|^2 T_s(\theta) + |e_p|^2 T_p(\theta) = T_{\text{avg}}(\theta(\mathbf{r})) + \frac{q(\mathbf{r})}{2} \Delta T(\theta(\mathbf{r}))$$

Unpolarised baseline:

$$T_{\text{avg}}(\theta) = \frac{T_s(\theta) + T_p(\theta)}{2}$$

TM excess:

$$\Delta T(\theta) = T_p(\theta) - T_s(\theta)$$

$\Delta T > 0$ below pseudo-Brewster angle;
 $\Delta T < 0$ above it.

Local TM excess parameter:

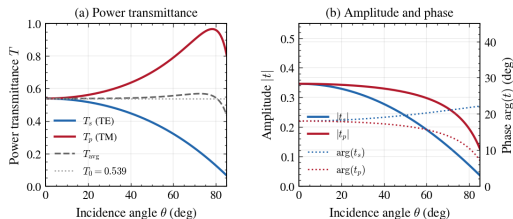
$$q(\mathbf{r}) = |e_p|^2 - |e_s|^2 \in [-1, 1]$$

- $q = +1$: pure TM
- $q = -1$: pure TE
- $q = 0$: unpolarised or circular

Two routes to geometric law:

1. $q = 0$: correction vanishes
2. $T_{\text{avg}} \approx T_0$: pseudo-Brewster

Pseudo-Brewster: mechanism and frequency dependence



T_s decreases monotonically.

T_p rises to pseudo-Brewster peak, then falls.

Average $T_{avg} \approx T_0$.

Azzam (2015): for $|\tilde{n}| > 2.5$, unpolarised reflectance deviates $< 1\%$ from normal incidence over $0-60^\circ$.

Tissue	$ \tilde{n} $	T_0	Max var.
Skin	4.84	0.54	$\leq 5.6\%$
Muscle	5.62	0.48	$\leq 4.8\%$
Fat	2.05	0.77	$\leq 8.2\%$
Water	6.62	0.45	$\leq 3.9\%$

Variation of T_{avg}/T_0 over $\theta \in [0^\circ, 75^\circ]$.

Exact at 40 GHz; conservative below;

3.5 % overestimate above 80 GHz.

Projected area, self-shadowing, and Cauchy

Total absorbed power:

$$P_{\text{abs}} = S_{\text{inc}} T_0 A_{\perp}(\hat{\mathbf{k}})$$

$A_{\perp}(\hat{\mathbf{k}}) = \int_{\Sigma_+} \cos \theta \, dA$: shadow area.

Exposure fraction (self-shadowing):

$$\eta(\mathbf{r}) = \frac{1}{\pi} \int_{S^2} \text{ReLU}[\hat{\mathbf{n}} \cdot (-\hat{\mathbf{k}})] O(\mathbf{r}, \hat{\mathbf{k}}) \, d\Omega$$

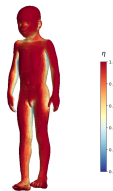
$O(\mathbf{r}, \hat{\mathbf{k}}) \in \{0, 1\}$: occlusion function.

$\eta \equiv 1$ convex; $\eta < 1$ concave.

$\equiv$ ambient occlusion (GPU graphics).

Absorption area: $A_{\text{ab}} = \int \eta(\mathbf{r}) \, dA$

Thelonious: $A_{\text{ab}}/A \approx 0.87$.

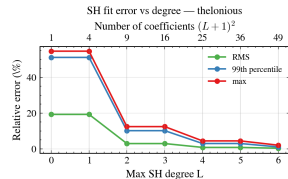
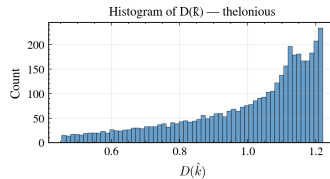
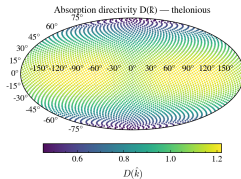


Generalised Cauchy

$$\langle P_{\text{abs}} \rangle = S_{\text{inc}} T_0 \frac{A_{\text{ab}}}{4}$$

Classical Cauchy ($\langle A_{\perp} \rangle = A/4$ for convex) is special case $\eta \equiv 1$.

Absorption directivity: detailed analysis



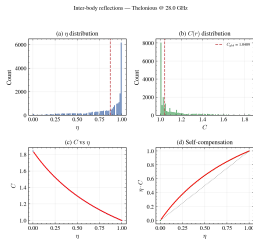
Mollweide projection

Histogram of $D(\hat{\mathbf{k}})$ SH fit error vs L

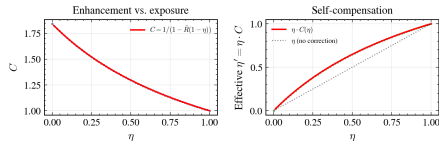
$$D(\hat{\mathbf{k}}) = 4A_{\perp}(\hat{\mathbf{k}})/A_{\text{ab}} \Rightarrow P_{\text{abs}}(\hat{\mathbf{k}}) = S_{\text{inc}} T_0 (A_{\text{ab}}/4) D(\hat{\mathbf{k}})$$

- Range on Thelonious: $D \in [0.8, 2.0]$. Front/back illumination highest; lateral lowest.
- $D(\hat{\mathbf{k}})$ smooth on S^2 : SH expansion $L \leq 6$ (49 coefficients) gives RMS < 1 %.
- Hemisphere bound: $D \leq 2$ (used for worst-case compliance).

Inter-body reflections



Thelonious @ 28.0 GHz — Inter-body reflections



- Reflected power from one body part illuminating another: sub-percent effect on body-averaged absorption
- Self-compensation mechanism: high-reflectivity directions also have low incidence cosine
- Justifies treating the body as a single-interaction absorber

Whole-body SAR compliance

Whole-body SAR:

$$\text{SAR}_{\text{wb}} = \frac{P_{\text{abs}}}{m} = \frac{S_{\text{inc}} T_0 A_{\perp}}{m}$$

Using hemisphere bound $D \leq 2$:

$$A_{\perp} \leq \frac{A_{\text{ab}}}{4} \cdot 2 = \frac{A_{\text{ab}}}{2}$$

Compliance criterion

$$S_{\text{inc}} < \frac{0.16 m}{T_0 A_{\text{ab}}}$$

Factor $0.16 = 2 \times \text{SAR}_{\text{limit}}/4$ where
 $\text{SAR}_{\text{limit}} = 0.08 \text{ W/kg}$ (ICNIRP).

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Body-size dependence:

Du Bois formula:

$$\text{BSA} = 0.007184 W^{0.425} H^{0.725}$$

The ratio m/A_{ab} scales as:

$$\frac{m}{A_{\text{ab}}} \propto \text{BMI}^{0.575} h^{0.425}$$

Children are the binding case for population-level limits:

- Lower m/A_{ab} ratio
- Compliance threshold $\sim 2\times$ lower than adults

Computational framework: ReLU network

Multi-source matrix formulation:

$$\mathbf{S}_{ab} = T_0 \text{ReLU}(\mathbf{N}\mathbf{K}^\top) \mathbf{s}$$

Neural network interpretation:

- $\mathbf{N} \in \mathbb{R}^{P \times 3}$: surface normals
- $\mathbf{K} \in \mathbb{R}^{N \times 3}$: path directions (negated)
- $\mathbf{s} \in \mathbb{R}^N$: path power densities
- $\mathbf{N}\mathbf{K}^\top$: “weight matrix” (dot products)
- ReLU: activation function (exact, not approximate)

Hierarchical computation:

- Full spatial map: $O(P \times N)$
- Total power: $O(N)$ (aggregate)
- Mean SAR: $O(1)$ after precomputation

Differentiable:

Gradients of SAR_{wb} w.r.t. antenna positions propagate through the chain via backpropagation.

Near-field extension

For a point source at distance $d > 3\lambda$:

- Law holds pointwise with spatially varying $S_{\text{inc}}(\mathbf{r})$ and $\hat{\mathbf{k}}(\mathbf{r})$
- Integrating over illuminated surface:

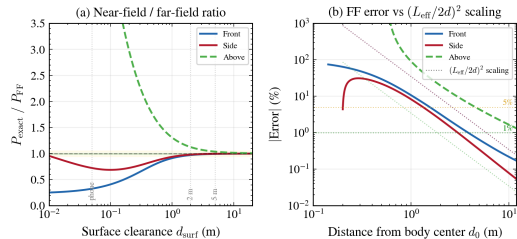
Solid-angle formula

$$P_{\text{abs}} = P_t T_0 \frac{\Omega_{\text{body}}}{4\pi}$$

Ω_{body} : solid angle subtended by the visible body surface from the source.

This is the emissive view-factor formula of radiative heat transfer, with T_0 replacing emissivity.

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Top: near-field vs far-field ratio.

Bottom: error vs distance showing $(L_{\text{eff}}/2d)^2$ scaling.

Higher-order corrections

Hierarchy of approximations:

Level	Approximation	Error
Exact law	$S_{ab} = S_{inc} T_{eff} \text{ReLU}(\mu)$	0 (by construction)
Geometric law	$S_{ab} \approx S_{inc} T_0 \text{ReLU}(\mu)$	$\leq 8.2\%$ (Fresnel)
+ Curvature	Physical optics correction	$\sim 0.5\%$ extra
+ Diffraction	Smooth activation (GELU-like)	$\sim 10\%$ at shadow boundary

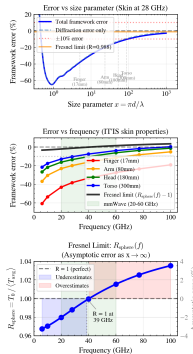
Curvature correction:

- Physical optics (PO) accounts for surface curvature
- Correction $\sim 0.5\%$ for most body regions
- Relevant only for small, highly curved features

Diffraction smoothing:

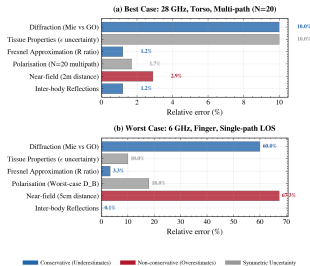
- Sharp ReLU $\rightarrow$ smooth activation near shadow boundary
- $\sim 10\%$ error for torso at 28 GHz
- Dominated by diffraction into geometric shadow
- Error is conservative (underestimates)

Mie theory validation: detailed analysis



Three panels: error vs size parameter (top), error vs frequency (middle), Fresnel asymptotic limit (bottom).
 Error $\sim 10\%$ at mmWave, dominated by diffraction. Torso at 28 GHz: $\sim 5\%$ (within tissue uncertainty).
 Conservative: underestimates absorption.

Error budget: detailed breakdown



Left: best case (28 GHz, multipath) — Fresnel $\sim 1\%$ integrated, diffraction averaged out.

Right: worst case (6 GHz, single-path LOS) — up to 15% for small body parts.

Tissue dielectric uncertainty: $\sim 20\%$ (dominant). Framework error $<$ tissue data uncertainty. Blue bars conservative.

3D absorbed power density on Thelonious



Front view



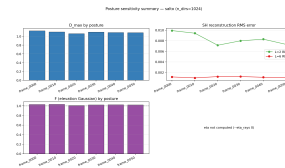
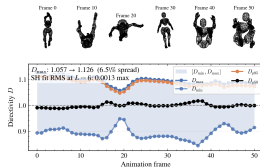
Isometric view

4 cm² averaged

Thelonious phantom, 28 GHz, frontal plane-wave illumination at ICNIRP reference level (10 W/m²).

- S_{ab} peaks on front-facing surfaces with highest $\cos \theta$
- Self-shadowed regions (armpits, inner legs): $S_{ab} \approx 0$
- 4 cm² spatial averaging (regulatory standard) smooths local peaks

Posture sensitivity analysis

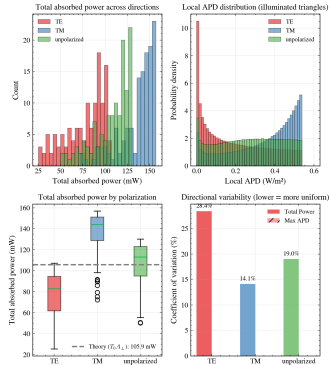


- RMS variation across postures: 6.5 %
- A_{ab}/A varies with arm position (changes self-shadowing)
- Arms raised: less self-shadowing, more absorption
- Arms at sides: armpits shadowed, less absorption

Posture variation (6.5 %) comparable to pseudo-Brewster error (5.6 %). Both smaller than tissue dielectric uncertainty (20 %).

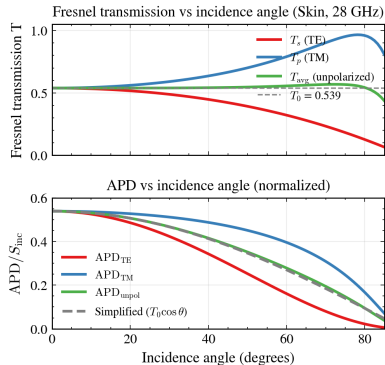
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APD directional analysis



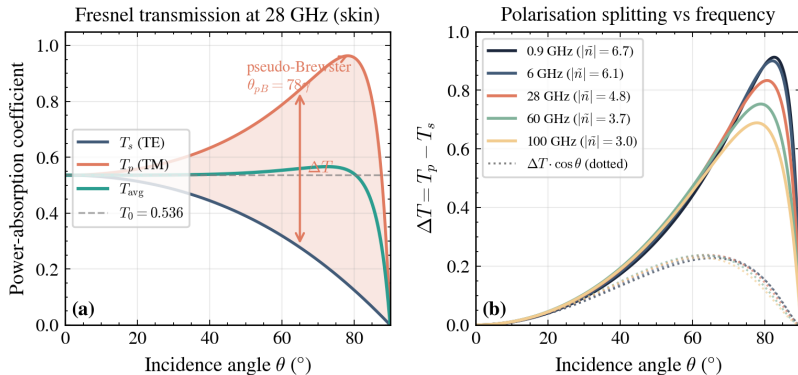
S_{ab} by polarisation: TE, TM, and unpolarised illumination compared across directions. Total absorbed power and directional variability metrics.

APD angle dependence



TE: monotonically decreasing with angle (always $\leq T_0$, conservative). TM: rises to pseudo-Brewster peak, then falls. Unpolarised average: near-constant (pseudo-Brewster compensation).

Polarisation splitting ΔT vs angle



$\Delta T = T_p - T_s$: peaks at pseudo-Brewster angle, drives polarisation sensitivity. For skin at 28 GHz: peak $\Delta T \approx 0.3$ near 60° .

Absorption Stokes vector derivation

Retain polarisation correction $q \neq 0$ in the exact law and integrate over the body:

Total absorbed power:

$$P_{\text{abs}} = \int_{\Sigma_+} S_{\text{inc}} T_{\text{eff}}(\mathbf{r}) \cos \theta \, dA$$

Substitute $T_{\text{eff}} = T_{\text{avg}} + \frac{q}{2} \Delta T$:

$$\begin{aligned} P_{\text{abs}} &= S_{\text{inc}} \int_{\Sigma_+} T_{\text{avg}} \cos \theta \, dA \\ &\quad + \frac{S_{\text{inc}}}{2} \int_{\Sigma_+} q(\mathbf{r}) \Delta T \cos \theta \, dA \end{aligned}$$

The local TM excess $q(\mathbf{r})$ depends on the incident Stokes parameters via the local orientation of the incidence plane.

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Result

$$P_{\text{abs}} = \mathbf{m}(\hat{\mathbf{k}}) \cdot \mathbf{s}_{\text{inc}}$$

$$\mathbf{s}_{\text{inc}} = (S_0, S_1, S_2, S_3)^\top \quad \mathbf{m}(\hat{\mathbf{k}}) = \begin{pmatrix} \tilde{A}_\perp \\ -\frac{1}{2}B_c \\ -\frac{1}{2}B_s \\ 0 \end{pmatrix}$$

$$\tilde{A}_\perp = \int T_{\text{avg}} \cos \theta \, dA$$

$$\tilde{B} = B_c + iB_s = \int \Delta T \cos \theta e^{2i\alpha} \, dA$$

Both precomputed from mesh.

Polarisation directivity D_B

The fractional polarisation correction is bounded by:

$$D_B(\hat{\mathbf{k}}) \equiv \frac{|\tilde{B}|}{2\tilde{A}_\perp}$$

Interpretation:

Maximum fractional deviation of P_{abs} from the unpolarised value, over all possible incident polarisation states.

Bounds:

- Sphere: $D_B = 0$
(polarisation-independent; proved in monograph)
- Cylinder at broadside: $D_B^{\text{cyl}} = 27.9\%$
(skin, 28 GHz)
- Thelonious (max over S^2): 16.0%

Why it matters for compliance:

The unpolarised geometric law is conservative for vertically polarised illumination:

- Horizontal limbs are purely TE
- $T_s(\theta) \leq T_0$ for all θ
- Simplified formula *overestimates* limb absorption

Worst-case (fully polarised, worst direction): +16 % on Thelonious.

Canonical shapes: sphere and cylinder

Sphere:

- Every incidence plane orientation is equally represented
- $\int e^{2i\alpha} dA = 0$ by symmetry
- $\tilde{B} = 0$, so $D_B = 0$
- Polarisation is *exactly* irrelevant
- Proof: rotational symmetry of the orientation integral

This also implies: a body composed of many randomly oriented convex patches approaches $D_B \rightarrow 0$.

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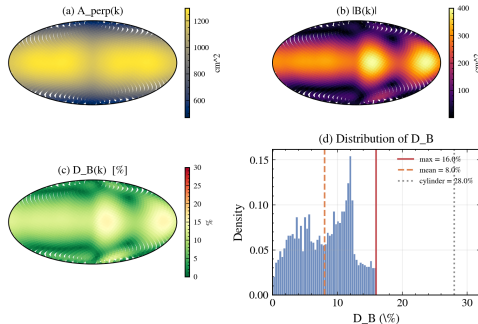
Infinite cylinder at broadside:

- All surface points share the same incidence plane
- $e^{2i\alpha} = 1$ everywhere
- $\tilde{B}$ is maximal relative to $\tilde{A}_\perp$
- Universal worst case: $D_B^{\text{cyl}} = 27.9\%$

For skin at 28 GHz. The cylinder bound is sharp: no body shape can exceed it (proved in monograph).

Human limbs approximate cylinders but at various orientations $\Rightarrow$ partial cancellation $\Rightarrow$ $D_B < 27.9\%$ on full body.

Polarisation response of Thelonious



Mollweide projection over all illumination directions.

Four panels: $\hat{A}_{\perp}$ (Fresnel-weighted projected area), $|B_c|$ and $|B_s|$ (polarisation correction components), D_B (fractional sensitivity).

Maximum $D_B = 16.0\%$ at broadside of torso (most cylinder-like direction).

Multipath convergence: detailed analysis

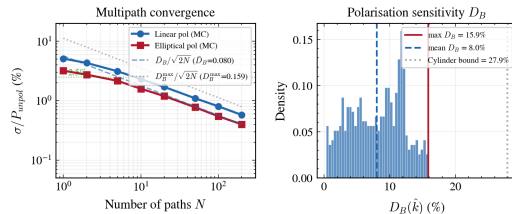
With N independent paths of random polarisation and direction:

$$\frac{\sigma_{\text{pol}}}{P_{\text{abs}}} = \frac{D_B}{\sqrt{2N}}$$

N	Std dev	vs pseudo-Brewster
1	up to 16 %	3× larger
5	≤ 5.1 %	comparable
10	≤ 3.6 %	smaller
20	≤ 2.5 %	half
50	≤ 1.6 %	$\ll$

Pseudo-Brewster error: ≤ 5.6 % for skin.

At $N = 10$: polarisation error already smaller.



Left: convergence with N .

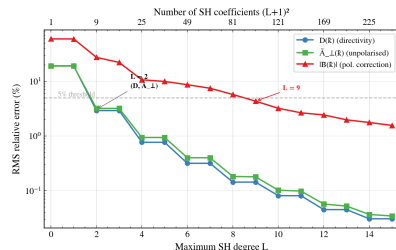
Right: histogram of D_B over directions.

Spherical harmonic compression

$\mathbf{m}(\hat{\mathbf{k}})$ varies smoothly over S^2 . SH expansion:

- $L = 6$: 49 coefficients per component
- $L = 9$: 100 coefficients

Compression: Full grid $(180 \times 360) = 64\,800$ values. SH $L=9$: 100 coeff. $\Rightarrow 648\times$. Enables precomputation of full polarisation response. With Stokes inner product: real-time dosimetry.



RMS error vs truncation order L . $L = 9$: error $< 5\%$.

Exact polarisation bounds

TE one-sided bound: For TE (s-polarised) illumination:

$$T_s(\theta) \leq T_0 \quad \forall \theta \in [0, \pi/2]$$

T_s is monotonically decreasing. $\Rightarrow$ The unpolarised geometric law *overestimates* absorption for purely TE illumination. This is the conservative direction for compliance.

Peak S_{ab} is polarisation-universal: At normal incidence ($\theta = 0$):

- $T_s(0) = T_p(0) = T_0$
- TE/TM distinction vanishes
- $T_{\text{eff}} = T_0$ regardless of polarisation

The peak S_{ab} (most directly facing point) is polarisation-independent. Polarisation matters only at oblique incidence, where S_{ab} is already reduced by $\cos \theta$. Combined effect: correction is bounded and conservative for compliance.

Exact averaged transmission $\bar{T}$

The flux-averaged transmission:

$$\bar{T}(f) = 2 \int_0^1 T_{\text{avg}}(\mu, f) \mu \, d\mu$$

Computed once per frequency from Fresnel coefficients.

Exact at any frequency

$$\langle P_{\text{abs}} \rangle = S_{\text{inc}} \bar{T} \frac{A_{\text{ab}}}{4}$$

$\bar{T} = T_0$ at 40 GHz (exact match).

$\bar{T} > T_0$ below 40 GHz:

T_0 -based formula underestimates (conservative).

$\bar{T} < T_0$ above 80 GHz:

f (GHz)	T_0	$\bar{T}$
0.9	0.63	0.72
2.4	0.57	0.61
6	0.52	0.55
10	0.53	0.54
28	0.54	0.54
40	0.53	0.53
60	0.50	0.50
100	0.45	0.43

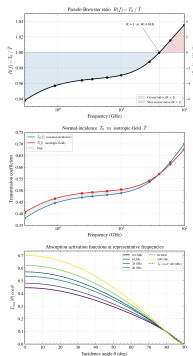
Values for skin

(IT'IS v5.0).

$T_0 \approx \bar{T}$ from 10 GHz;

diverge below 6 GHz (Fabry-Pérot).

Frequency overview: 100 MHz to 100 GHz



Pseudo-Brewster accuracy as a function of frequency.

Green band (10 GHz–60 GHz): geometric absorption law optimal (T_0 approximation within 5%).

Below 10 GHz: use $\bar{T}$ for total power (local map loses physical meaning below 6 GHz).

Above 80 GHz: T_0 overestimates by up to 3.5 % (non-conservative).

Validity regimes

	0.1 GHz–6 GHz	6 GHz–10 GHz	10 GHz–60 GHz	60 GHz–100 GHz
Local $S_{\text{ab}}(\mathbf{r})$ with T_0	No	Approx	Yes ($\leq 6\%$)	Yes ($\leq 8\%$)
Total P_{abs} with T_0	Conservative	Yes	Yes	Yes
Total P_{abs} with $\bar{T}$	Exact	Exact	Exact	Exact
Compliance formula	Conservative	Yes	Yes	Yes

Below 6 GHz: the local surface map $S_{\text{ab}}(\mathbf{r})$ loses physical meaning when multi-layer tissue resonances (Fabry-Pérot between skin, fat, muscle) dominate. Total absorbed power and compliance criteria remain valid everywhere via $\bar{T}$.

Setup and inherited assumptions

Antenna array and precoding:

- M antenna elements, precoding vector $\mathbf{x} \in \mathbb{C}^M$ with $\|\mathbf{x}\|^2 = P$
- N propagation paths from ray tracer or channel model
- Each path n carries polarisation-amplitude vector $\psi_n \in \mathbb{C}^3$

Inherited from Parts I–II:

- Locally flat surface at each body-surface point $\mathbf{r}$
- Opaque body ($|\tilde{n}| > 2.5$, skin-depth regime)
- Fresnel transmission at each surface point
- Depth-integrated Ohmic loss defines $S_{\text{ab}}(\mathbf{r})$

What changes in Part III:

- **Coherent summation of complex amplitudes** (not powers)
- Cross-terms $\psi_n^H \psi_{n'} x_{j(n)} x_{j(n')}^*$ can no longer be ignored
- Must retain exact complex Fresnel coefficients t_s, t_p (not $\sqrt{T_0}$)

Propagation paths

Each path $n = 1, \dots, N$ is characterised by a triplet:

Path descriptor

$$(j(n), \hat{\mathbf{k}}_n, \psi_n) \quad \begin{cases} j(n) \in \{1, \dots, M\} & \text{feeding antenna element} \\ \hat{\mathbf{k}}_n \in \mathbb{R}^3, \|\hat{\mathbf{k}}_n\| = 1 & \text{direction of arrival} \\ \psi_n \in \mathbb{C}^3 & \text{polarisation-amplitude vector} \end{cases}$$

ψ_n encodes all propagation interactions:

- Complex amplitude (including path loss and phase)
- Polarisation state (elliptical in general)
- Reflection, diffraction, scattering along the path

At body surface point $\mathbf{r}$:

- Local incidence angle: $\theta_n(\mathbf{r}) = \arccos(\hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}}_n))$
- Wave decomposes into TE/TM relative to local incidence plane
- TE/TM unit vectors $\mathbf{e}_{s,n}(\mathbf{r})$ and $\mathbf{e}_{p,n}(\mathbf{r})$ depend on $\hat{\mathbf{k}}_n$ and $\hat{\mathbf{n}}(\mathbf{r})$

Field channel matrix

$$\mathbf{E}(\mathbf{r}) = \sum_{n=1}^N \psi_n x_{j(n)}$$

Signal channel (scalar at UE):

$$\mathbf{h} \in \mathbb{C}^M, \quad y = \mathbf{h}^\top \mathbf{x}$$

Each element of $\mathbf{h}$: scalar sum of path contributions arriving at UE.

Exposure channel (vector at body):

$$\tilde{\mathbf{G}}(\mathbf{r}) \in \mathbb{C}^{3 \times M}$$

Column j of $\tilde{\mathbf{G}}(\mathbf{r})$:

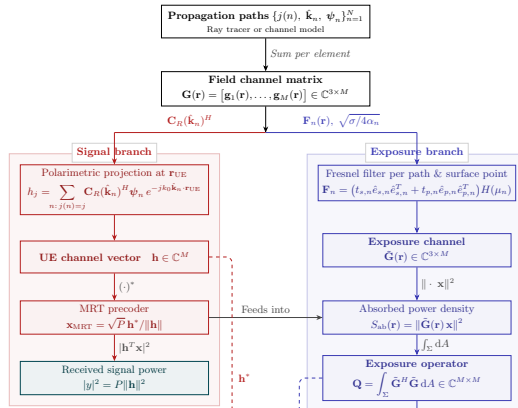
$$[\tilde{\mathbf{G}}(\mathbf{r})]_j = \sum_{n: j(n)=j} F_n(\mathbf{r}) \psi_n$$

Key structural difference

Signal sums scalars $\rightarrow \mathbf{h}^\top \mathbf{x} \in \mathbb{C}$

Exposure sums vectors $\rightarrow \tilde{\mathbf{G}}(\mathbf{r}) \mathbf{x} \in \mathbb{C}^3$

Signal vs exposure: propagation to field channel

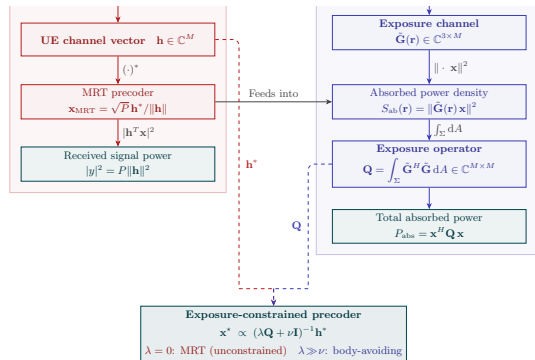


Top half: propagation paths $\rightarrow$ field channel matrix $\rightarrow$ signal/exposure branch split.

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Signal vs exposure: branches to output



$\mathbf{C}_n^H$ collapses each path to a *scalar* at $\mathbf{r}_{\text{UE}}$.
 $\mathbf{F}_n(\mathbf{r})$ retains the full *vector* structure over the body surface Σ .

Bottom half: signal $\rightarrow y = \mathbf{h}^T \mathbf{x}$; exposure $\rightarrow P_{\text{abs}} = \mathbf{x}^H \mathbf{Q} \mathbf{x}$. Precoder balances both.

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Signal vs exposure: key differences

Signal branch:

1. Propagation paths ψ_n
2. Scalar collapse at UE antenna
3. Received signal: $y = \mathbf{h}^\top \mathbf{x}$

UE antenna selects one polarisation component $\rightarrow$ scalar per path.

Both branches share the same propagation paths and precoding vector $\mathbf{x}$.
The structural difference is why signal focus $\neq$ body hotspot.

Exposure branch:

1. Same propagation paths ψ_n
2. Fresnel filter $F_n(\mathbf{r})$ at body surface
3. Vector field: $\|\tilde{\mathbf{G}}(\mathbf{r}) \mathbf{x}\|^2$

Fresnel filter retains polarisation and angular dependence $\rightarrow 3 \times M$ matrix per surface point.

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Fresnel transmission operator

For path n at surface point $\mathbf{r}$:

Fresnel operator

$$F_n(\mathbf{r}) = \left(t_{s,n} \mathbf{e}_{s,n} \mathbf{e}_{s,n}^\top + t_{p,n} \mathbf{e}_{p,n} \mathbf{e}_{p,n}^\top \right) H(\mu_n(\mathbf{r}))$$

Components:

- $t_{s,n}, t_{p,n}$: complex Fresnel transmission coeff.
- $\mathbf{e}_{s,n}, \mathbf{e}_{p,n}$: TE/TM unit vectors
- $H(\mu_n)$: Heaviside step (visibility gate)
- $\mu_n(\mathbf{r}) = \hat{\mathbf{n}}(\mathbf{r}) \cdot (-\hat{\mathbf{k}}_n)$

F_n : rank-2 if front-facing ($\mu_n > 0$), rank-0 if back-facing

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Why exact t_s, t_p matter:

- Cross-terms depend on phase of t_s, t_p
- $\sqrt{T_0}$ replacement destroys phases
- Cross-term error: unbounded
- Self-terms: $|t_s|^2 = T_s, |t_p|^2 = T_p$

Incoherent (Part I): only $|t|^2$ matters. Coherent (Part III): full complex t required.

Coherent depth integral and Approximation 1

Absorbed power density = Ohmic loss integrated over depth z :

$$S_{\text{ab}}(\mathbf{r}) = \int_0^\infty \frac{\sigma}{2} |\mathbf{E}_t(\mathbf{r}, z)|^2 dz$$

Expand $|\mathbf{E}_t|^2$: self-terms + cross-terms weighted by depth coupling:

$$\Lambda_{nn'}(\mathbf{r}) = \int_0^\infty \frac{\sigma}{2} e^{-\alpha_n z} e^{-\alpha_{n'}^* z} dz \quad (\text{complex decay rates } \alpha_n, \alpha_{n'})$$

Approximation 1: TM direction replacement

Replace refracted TM direction by incident direction inside the medium.

- Error: $O(1/|\tilde{n}|^2) \leq 4\%$ on cross-terms only
- Self-terms: exact (no approximation needed)

Physical reason: for $|\tilde{n}| \gg 1$, Snell's law gives refracted angle $\theta_t \approx \sin \theta_i / |\tilde{n}| \ll 1$. All refracted rays propagate nearly normal to the surface. At 28 GHz with $|\tilde{n}| \approx 5.5$: correction $\leq 3.3\%$ on cross-terms.

Universal depth coupling (Approximation 2)

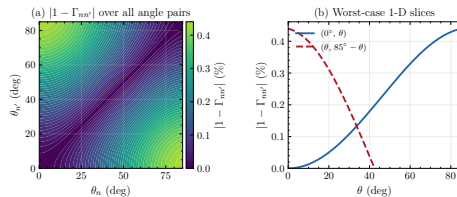
Normalised depth coupling:

$$\Gamma_{nn'} = \frac{\Lambda_{nn'}}{\sqrt{\Lambda_{nn} \Lambda_{n'n'}}$$

Approximation 2

$$\Gamma_{nn'} \approx 1 \quad \forall n, n'$$

Validation: exact to 0.44 % at 28 GHz, validated over all $(\theta_n, \theta_{n'})$. **Physical reason:** all paths decay with the same skin depth δ .



$\Gamma_{nn'}$ vs angle pairs. Deviation < 0.5 %.

Combined error budget:

Source	Max error
Approx. 1 (TM direction)	$\leq 4 \%$
Approx. 2 (depth coupling)	$\leq 0.44 \%$
Total on cross-terms	$\leq 5 \%$

Main theorem: coherent absorption law

Under Approximations 1 and 2, the double sum collapses to a squared norm:

Coherent absorption law

$$S_{\text{ab}}(\mathbf{r}) = \|\tilde{\mathbf{G}}(\mathbf{r}) \mathbf{x}\|^2, \quad \tilde{\mathbf{G}}(\mathbf{r}) \in \mathbb{C}^{3 \times M}$$

Column j : $[\tilde{\mathbf{G}}(\mathbf{r})]_j = \sum_{n: j(n)=j} \sqrt{\Lambda_{nn}(\mathbf{r})} F_n(\mathbf{r}) \psi_n$

Incoherent limit:

- Phases of ψ_n random
- Cross-terms $\rightarrow 0$
- Reduces to Part I formula

Coherent peak:

- All N paths co-phase at $\mathbf{r}$
- $S_{\text{ab}} \propto N^2$ (vs N incoherent)
- Factor N enhancement
- Precoder focuses energy on body

Approximation error $\leq 5\%$ on cross-terms; self-terms exact.

Exposure operator properties

Exposure operator

$$\mathbf{Q} = \int_{\Sigma} \tilde{\mathbf{G}}(\mathbf{r})^H \tilde{\mathbf{G}}(\mathbf{r}) dA \in \mathbb{C}^{M \times M}$$

Properties: Hermitian PSD ($\mathbf{Q} = \mathbf{Q}^H$, $\mathbf{x}^H \mathbf{Q} \mathbf{x} \geq 0$). Rank $\leq 3 \times |\text{illuminated surface pts}|$; full rank when $M \ll \text{surface points}$.

Eigendecomposition: $\mathbf{Q} = \sum_{k=1}^M \lambda_k \mathbf{v}_k \mathbf{v}_k^H$, $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$

Quantity	Meaning	Role
$\mathbf{v}_1$	Dominant eigenvector	Worst-case precoder
$\lambda_{\max}$	Largest eigenvalue	Max absorption / unit TX power
$\text{tr}(\mathbf{Q})$	Trace	Isotropic-TX expected absorption
$\mathbf{v}_k, \lambda_k$	k -th eigenpair	k -th exposure mode

Each $\mathbf{v}_k$ defines a distinct spatial absorption pattern on the body surface.

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Exposure-signal alignment ρ : derivation

MRT precoder: $\mathbf{x}_{\text{MRT}} = \sqrt{P} \mathbf{h}^* / \|\mathbf{h}\|$

Absorbed power under MRT:

$$P_{\text{abs}}^{\text{MRT}} = \mathbf{x}_{\text{MRT}}^H \mathbf{Q} \mathbf{x}_{\text{MRT}} = P \frac{\mathbf{h}^\top \mathbf{Q} \mathbf{h}^*}{\|\mathbf{h}\|^2}$$

Exposure-signal alignment

$$\rho \equiv \frac{\mathbf{h}^\top \mathbf{Q} \mathbf{h}^*}{\|\mathbf{h}\|^2 \lambda_{\max}(\mathbf{Q})} \in [0, 1]$$

Bound follows from the Rayleigh quotient: $\mathbf{h}^\top \mathbf{Q} \mathbf{h}^* / \|\mathbf{h}\|^2 \leq \lambda_{\max}(\mathbf{Q})$.

Three reasons signal focus $\neq$ body hotspot ($\rho < 1$): (1) Polarisation mismatch body vs UE; (2) Grazing paths: $F_n \rightarrow 0$ but still contribute to signal; (3) Phase divergence between UE and body decorrelates $\mathbf{h}$ from $\mathbf{Q}$.

$\rho = 1$: MRT $\equiv$ worst-case precoder. $\rho = 0$: MRT in null space of $\mathbf{Q}$.

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QCQP formulation and solution

Problem: maximise signal power subject to absorption and power limits:

$$\max_{\mathbf{x}} |\mathbf{h}^\top \mathbf{x}|^2 \quad \text{s.t.} \quad \mathbf{x}^H \mathbf{Q} \mathbf{x} \leq P_{\text{abs}}^{\max}, \quad \|\mathbf{x}\|^2 \leq P$$

This is a QCQP. S-procedure: two quadratic constraints with PSD matrices $\Rightarrow$ strong duality.

Optimal precoder (KKT solution)

$$\mathbf{x}^* = \sqrt{P} \frac{(\lambda \mathbf{Q} + \nu \mathbf{I})^{-1} \mathbf{h}^*}{\|(\lambda \mathbf{Q} + \nu \mathbf{I})^{-1} \mathbf{h}^*\|}$$

$\lambda, \nu \geq 0$: dual variables found by bisection.

Regime	Condition	Behaviour
Unconstrained	$\lambda = 0$	MRT: $\mathbf{x}^* \propto \mathbf{h}^*$
Body-avoiding	$\lambda \gg \nu$	Suppresses high- λ_k eigenvectors of $\mathbf{Q}$
Natural protection	$\rho \ll 1$	MRT satisfies constraint; minimal signal loss

Same structure as Ying

et al. 2015, but $\mathbf{Q}$ derived from Fresnel physics, not FDTD-calibrated.

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Computational cost

Full FDTD baseline:

- $\sim 10^{12}$ mesh cells at 28 GHz
- Weeks per simulation per precoder
- Must re-run for each $\mathbf{x}$
- Systematic campaign: M precoders $\times$ frequencies $\times$ phantoms

Framework pipeline:

1. Body mesh + ray tracer output (offline)
2. Construct $\mathbf{Q}$: $O(M \cdot N_{\text{surf}} \cdot N_{\text{paths}})$
3. Eigendecomposition: $O(M^3)$
4. Precoder computation: $O(M^2)$

Cost comparison:

Stage	FDTD	Framework
$\mathbf{Q}$ construction	weeks	seconds
Per-precoder eval	weeks	μs
Eigendecomposition	N/A	ms

Speedup

Total speedup $\approx 10^6 \times$

Key advantage: $\mathbf{Q}$ from body mesh + ray tracer. Precoder update at channel-coherence time (ms), not FDTD (weeks).